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ELECTRONIC DEVICE COMPRISING STRUCTURE FOR DISPERSING HEAT

Abstract

An electronic device according to one embodiment may comprise a printed circuit board, a processor on the printed circuit board, a vapor chamber which is spaced apart from the printed circuit board, and of which at least a portion is disposed on the processor, and a metal plate on the printed circuit board, including a seating groove accommodating at least a portion of the vapor chamber. The vapor chamber can include a plurality of pillars having flat ends. The plurality of pillars can include a first set of pillars having a first pattern included in a portion of the vapor chamber, and a second set of pillars having a second pattern included in a different portion of the vapor chamber, the second pattern being different from the first pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation application, claiming priority under 35 U.S.C. § 365(c), of an International application No. PCT/KR2023/017696, filed on Nov. 6, 2023, which is based on and claims the benefit of a Korean patent application number 10-2023-0009085, filed on Jan. 20, 2023, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2023-0013279, filed on Jan. 31, 2023, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND
1. Field
[0002] Embodiments to be described later relate to an electronic device including a structure for dispersing heat.
2. Description of Related Art
[0003] An electronic device may include various electronic components in order to meet user needs. The electronic device may be miniaturized so that it may be worn by the user or carried by the user. As the electronic components disposed in the electronic device perform an operation for responding to a user's request, heat may be generated in the miniaturized electronic device.
SUMMARY
[0004] According to an embodiment, an electronic device may comprise a printed circuit board, a processor on the printed circuit board, and a vapor chamber spaced apart from the printed circuit board and at least partially disposed over the processor. The electronic device may comprise a metal plate, disposed on the printed circuit board, including a seating groove in which at least a portion of the vapor chamber **400** is accommodated and including a first portion on the processor for transferring heat emitted from the processor to the vapor chamber and a second portion extending from the first portion and having a slit formed therein. The vapor chamber may include a first plate including a base supported by the first portion of the seating groove and having at least a portion disposed on the slit, and a sidewall extending from a periphery of the base toward an outside of the seating groove. The vapor chamber may include a second plate contacting the sidewall and including a first region disposed on the first portion, a second region disposed on the slit of the second portion, a third region extending from the first region to the second region and located on the boundary between the first portion and the second portion, and a plurality of pillars protruding toward the base and including an end having flat shape at a position adjacent to the base. The plurality of pillars may include a first set of pillars having a first pattern in the first region and the second region, and a second set of pillars having a second pattern different from the first pattern in the third region.
[0005] According to an embodiment, an electronic device may comprise a printed circuit board, a processor on the printed circuit board, and a vapor chamber spaced apart from the printed circuit board and at least partially disposed over the processor. The electronic device may comprise a

metal plate, disposed on the printed circuit board, including a seating groove in which at least a portion of the vapor chamber is accommodated and including a first portion on the processor for transferring heat emitted from the processor to the vapor chamber and a second portion extending from the first portion and having a slit formed therein. The vapor chamber may include a first plate including a base supported by the first portion of the seating groove and having at least a portion disposed on the slit, and a sidewall extending from a periphery of the base toward an outside of the seating groove. The vapor chamber may include a second plate contacting the sidewall and including a first region disposed on the first portion, a second region disposed on the slit of the second portion, a third region extending from the first region to the second region and located on the boundary between the first portion and the second portion, and a plurality of pillars protruding toward the base and including an end having flat shape at a position adjacent to the base. The vapor chamber may further include at least one of stainless steel and titanium. The plurality of pillars may include a first set of pillars having a first pattern in the first region and the second region, and a second set of pillars having a second pattern different from the first pattern in the third region. A rigidity of the third region may be greater than a rigidity of the first region and a rigidity of the second region.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** is a block diagram of an electronic device in a network environment according to an embodiment.

[0007] FIG. **2** is a diagram illustrating an electronic device according to an embodiment.

[0008] FIG. **3** is an exploded perspective view of an electronic device according to an embodiment.

[0009] FIG. **4**A illustrates an exemplary electronic device.

[0010] FIG. **4**B is a cross-sectional view of an exemplary vapor chamber cut along line A-A′ of FIG. **4**A.

[0011] FIGS. **5**A, **5**B, and **5**C illustrate a portion of an exemplary vapor chamber.

[0012] FIGS. **6**A and **6**B illustrate a portion of an exemplary vapor chamber.

[0013] FIG. **7**A illustrates the inside of an exemplary electronic device in which a vapor chamber is disposed.

[0014] FIG. **7**B is a cross-sectional view of an exemplary electronic device cut along line B-B′ of FIG. **7**A.

[0015] FIG. **7**C is a cross-sectional view of an exemplary electronic device cut along line C-C′ of FIG. **7**A.

DETAILED DESCRIPTION

[0016] FIG. **1** is a block diagram illustrating an electronic device **101** in a network environment **100** according to various embodiments.

[0017] Referring to FIG. **1**, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

[0018] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0019] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

[0020] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0021] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0022] The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).

[0023] The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0024] The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.

[0025] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0026] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0027] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0028] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0029] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0030] The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0031] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0032] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0033] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable

independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.
[0034] The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.
[0035] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment, the antenna module **197** may include an antenna including a radiating element composed of a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.
[0036] According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface

(e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

[0037] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0038] According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

[0039] FIG. **2** is a diagram illustrating an electronic device according to an embodiment.

[0040] Referring to FIG. **2**, an electronic device **101** according to an embodiment may include a housing **210** forming an exterior of the electronic device **101**. For example, the housing **210** may include a front surface **200**A, a rear surface **200**B, and a side surface **200**C surrounding a space between the front surface **200**A and the rear surface **200**B. According to an embodiment, the housing **210** may refer to a structure forming at least a portion of the front surface **200**A, the rear surface **200**B, and/or the side surface **200**C.

[0041] The electronic device **101** according to an embodiment may include a substantially transparent front plate **202**. According to an embodiment, the front plate **202** may form at least a portion of the front surface **200**A. According to an embodiment, the front plate **202** may include, for example, a glass plate including various coating layers or a polymer plate, but the disclosure is not limited thereto.

[0042] The electronic device **101** according to an embodiment may include a substantially opaque rear plate **211**. According to an embodiment, the rear plate **211** may form at least a portion of the rear surface **200**B. According to an embodiment, the rear plate **211** may be formed of coated or colored glass, ceramic, polymer, metal (e.g., aluminum, stainless steel (STS), or magnesium), or a combination of at least two of the above materials.

[0043] The electronic device **101** according to an embodiment may include a side structure (or side member) **218**. According to an embodiment, the side structure **218** may be coupled to the front plate **202** and/or the rear plate **211** to form at least a portion of the side surface **200**C of the electronic device **101**. For example, the side structure **218** may form all of the side surface **200**C of the electronic device **101**, and for another example, the side structure **218** may form the side surface **200**C of the electronic device **101** together with the front plate **202** and/or the rear plate **211**.

[0044] Unlike the illustrated embodiment, in the case that the side surface **200**C of the electronic device **101** is partially formed by the front plate **202** and/or the rear plate **211**, the front plate **202** and/or the rear plate **211** may include a region that is bent from its periphery toward the rear plate **211** and/or the front plate **202** and seamlessly extends. The extended region of the front plate **202** and/or the rear plate **211** may be positioned at both ends of, for example, a long edge of the electronic device **101**, but the disclosure is not limited to the above-described examples.

[0045] According to an embodiment, the side structure **218** may include a metal and/or a polymer. According to an embodiment, the rear plate **211** and the side structure **218** may be integrally formed and may include the same material (e.g., a metal material such as aluminum), but the disclosure are not limited thereto. For example, the rear plate **211** and the side structure **218** may be formed in separate configurations and/or may include different materials.

[0046] According to an embodiment, the electronic device **101** may include at least one of a display **201**, an audio module **203**, **204** and **207**, a sensor module (not shown), a camera module **205**, **212**, **213**, a key input device **217**, a light emitting device (not shown), and/or a connector hole **208**. According to another embodiment, the electronic device **101** may omit at least one of the components (e.g., a key input device **217** or a light emitting device (not shown)), or may further include another component.

[0047] According to an embodiment, the display **201** may be visually exposed through a substantial portion of the front plate **202**. For example, at least a portion of the display **201** may be visible through the front plate **202** forming the front surface **200**A. According to an embodiment, the display **201** may be disposed on the second surface of the front plate **202**.

[0048] According to an embodiment, the appearance of the display **201** may be formed substantially the same as the appearance of the front plate **202** adjacent to the display **201**. According to an embodiment, in order to expand the area in which the display **201** is visually exposed, the distance between the outside of the display **201** and the outside of the front plate **202** may be formed to be generally the same.

[0049] According to an embodiment, the display **201** (or the front surface **200**A of the electronic device **101**) may include a screen display area **201**A. According to an embodiment, the display **201** may provide visual information to a user through the screen display area **201**A. In the illustrated embodiment, when the front surface **200**A is viewed from the front, it is illustrated that the screen display area **201**A is spaced apart from the outside of the front surface **200**A and is positioned inside the front surface **200**A, but the disclosure is not limited thereto. In another embodiment, when the front surface **200**A is viewed from the front, at least a portion of the periphery of the screen display area **201**A may substantially coincide with the periphery of the front surface **200**A (or the front plate **202**).

[0050] According to an embodiment, the screen display area **201**A may include a sensing area **201**B configured to obtain biometric information of a user. Here, the meaning of “the screen display area **201**A including the sensing area **201**B” may be understood to mean that at least a portion of the sensing area **201**B may be overlapped on the screen display area **201**A. For example, the sensing area **201**B, like other areas of the screen display area **201**A, may refer to an area in which visual information may be displayed by the display **201** and additionally biometric information (e.g., fingerprint) of a user may be obtained. In another embodiment, the sensing area **201**B may be formed in the key input device **217**.

[0051] According to an embodiment, the display **201** may include an area in which the first camera module **205** is positioned. According to an embodiment, an opening may be formed in the area of the display **201**, and the first camera module **205** (e.g., a punch hole camera) may be at least partially disposed in the opening to face the front surface **200**A. In such a case, the screen display area **201**A may surround at least a portion of the periphery of the opening. According to an embodiment, the first camera module **205** (e.g., an under display camera (UDC)) may be disposed under the display **201** to overlap the area of the display **201**. In this case, the display **201** may provide visual information to the user through the area, and additionally, the first camera module **205** may obtain an image corresponding to a direction facing the front surface **200**A through the area of the display **201**.

[0052] According to an embodiment, the display **201** may be coupled to or disposed adjacent to a touch sensing circuit, a pressure sensor capable of measuring the intensity (pressure) of the touch, and/or a digitizer that detects a magnetic field type stylus pen.

[0053] According to an embodiment, the audio modules **203**, **204** and **207** may include microphone holes **203** and **204**, and a speaker hole **207**.

[0054] According to an embodiment, the microphone holes **203** and **204** may include a first microphone hole **203** formed in a partial area of the side surface **200**C and a second microphone hole **204** formed in a partial area of the rear surface **200**B. A microphone (not shown) for obtaining an external sound may be disposed inside the microphone holes **203** and **204**. The microphone may include a plurality of microphones to detect the direction of sound.

[0055] According to an embodiment, the second microphone hole **204** formed in a partial area of the rear surface **200**B may be disposed adjacent to the camera modules **205**, **212** and **213**. For example, the second microphone hole **204** may obtain sound according to operations of the camera modules **205**, **212**, and **213**. However, the disclosure is not limited thereto.

[0056] According to an embodiment, the speaker hole **207** may include an external speaker hole **207** and a receiver hole (not illustrated) for a call. The external speaker hole **207** may be formed on a portion of the side surface **200**C of the electronic device **101**. According to an embodiment, the external speaker hole **207** may be implemented as a single hole together with the microphone hole **203**. Although not illustrated, a receiver hole (not shown) for a call may be formed on another portion of the side surface **200**C. For example, the receiver hole for a call may be formed on the opposite side of the external speaker hole **207** on the side surface **200**C. For example, with respect to the illustration of FIG. **2**, the external speaker hole **207** may be formed on the side surface **200**C corresponding to the lower end of the electronic device **101**, and the receiver hole for a call may be formed on the side surface **200**C corresponding to the upper end of the electronic device **101**. However, the disclosure is not limited thereto, and according to an embodiment, the receiver hole for a call may be formed at a position other than the side surface **200**C. For example, the receiver hole for a call may be formed by a space spaced apart between the front plate **202** (or display **201**) and the side bezel structure **218**.

[0057] According to an embodiment, the electronic device **101** may include at least one speaker (not illustrated) configured to output sound to the outside of the housing **210** through an external speaker hole **207** and/or a receiver hole (not shown) for a call.

[0058] According to an embodiment, the sensor module (not shown) may generate an electrical signal or data value corresponding to an internal operating state or an external environmental state of the electronic device **101**. For example, the sensor module may include at least one of a proximity sensor, an HRM sensor, a fingerprint sensor, a gesture sensor, a gyro sensor, a pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, and an illumination sensor.

[0059] According to an embodiment, the camera modules **205**, **212** and **213** may include a first camera module **205** disposed to face the front surface **200**A of the electronic device **101**, a second camera module **212** disposed to face the rear surface **200**B, and a flash **213**.

[0060] According to an embodiment, the second camera module **212** may include a plurality of cameras (e.g., a dual camera, a triple camera, or a quad camera). However, the second camera module **212** is not necessarily limited to including a plurality of cameras, and may include one camera.

[0061] According to an embodiment, the first camera module **205** and the second camera module

212 may include one or a plurality of lenses, an image sensor, and/or an image signal processor.

[0062] According to an embodiment, the flash **213** may include, for example, a light emitting diode or a xenon lamp. According to an embodiment, two or more lenses (infrared camera, wide-angle and telephoto lens) and image sensors may be disposed on one side of the electronic device **101**.

[0063] According to an embodiment, the key input device **217** (e.g., an input module **150** of FIG. **1**) may be disposed on the side surface **200**C of the electronic device **101**. According to an embodiment, the electronic device **101** may not include some or all of the key input devices **217**, and the key input device **217** not included therein may be implemented on the display **201** in another form such as a soft key.

[0064] According to an embodiment, the connector hole **208** may be formed on the side surface **200**C of the electronic device **101** to accommodate the connector of the external device. A connection terminal electrically connected to the connector of the external device may be disposed in the connector hole **208**. The electronic device **101** according to an embodiment may include an interface module for processing electrical signals transmitted and received through the connection terminal.

[0065] According to an embodiment, the electronic device **101** may include a light emitting device (not shown). For example, the light emitting device (not shown) may be disposed on the front surface **200**A of the housing **210**. The light emitting device (not shown) may provide state information of the electronic device **101** in a form of light. In another embodiment, the light emitting device (not shown) may provide a light source when the first camera module **205** is operated. For example, the light emitting device (not shown) may include an LED, an IR LED, and/or a xenon lamp.

[0066] FIG. **3** is an exploded perspective view of an electronic device according to an embodiment.

[0067] Hereinafter, overlapping descriptions of components having the same reference numerals as those of the above-described components will be omitted.

[0068] Referring to FIG. **3**, the electronic device **101** according to an embodiment may include a frame structure **240**, a first printed circuit board **250**, a second printed circuit board **252**, a cover plate **260**, and a battery **270**.

[0069] According to an embodiment, the frame structure **240** may include a sidewall **241** forming an exterior (e.g., the third surface **200**C of FIG. **2**) of the electronic device **101** and a support portion **243** extending inward from the sidewall **241**. According to an embodiment, the frame structure **240** may be disposed between the display **201** and the rear plate **211**. According to an embodiment, the sidewall **241** of the frame structure **240** may surround a space between the rear plate **211** and the front plate **202** (and/or the display **201**), and the support portion **243** of the frame structure **240** may extend from the sidewall **241** within the space.

[0070] According to an embodiment, the frame structure **240** may support or accommodate other components included in the electronic device **101**. For example, the display **201** may be disposed on one surface of the frame structure **240** facing one direction (e.g., the +z direction), and the display **201** may be supported by the support portion **243** of the frame structure **240**. For another example, a first printed circuit board **250**, a second printed circuit board **252**, a battery **270**, and a rear camera **212** may be disposed on the other surface facing a direction opposite to the one direction (e.g., the −z direction) of the frame structure **240**. The first printed circuit board **250**, the second printed circuit board **252**, the battery **270**, and the rear camera **212** may be mounted on a recess defined by the sidewall **241** and/or the support portion **243** of the frame structure **240**.

[0071] According to an embodiment, the first printed circuit board **250**, the second printed circuit board **252**, and the battery **270** may be coupled to the frame structure **240**, respectively. For example, the first printed circuit board **250** and the second printed circuit board **252** may be fixedly disposed in the frame structure **240** through a coupling member such as a screw. For example, the battery **270** may be fixedly disposed on the frame structure **240** through an adhesive member (e.g., a double-sided tape). However, it is not limited by the above-described example.

[0072] According to an embodiment, a cover plate **260** may be disposed between the first printed circuit board **250** and the rear plate **211**. According to an embodiment, the cover plate **260** may be disposed on the first printed circuit board **250**. For example, the cover plate **260** may be disposed on a surface facing the −z direction of the first printed circuit board **250**.
[0073] According to an embodiment, the cover plate **260** may at least partially overlap the first printed circuit board **250** with respect to the z-axis. According to an embodiment, the cover plate **260** may cover at least a partial area of the first printed circuit board **250**. Through this, the cover plate **260** may protect the first printed circuit board **250** from physical impact or prevent the connector coupled to the first printed circuit board **250** from being separated.
[0074] According to an embodiment, the cover plate **260** may be fixedly disposed on the first printed circuit board **250** through a coupling member (e.g., a screw), or may be coupled to the frame structure **240** together with the first printed circuit board **250** through the coupling member.
[0075] According to an embodiment, the display **201** may be disposed between the frame structure **240** and the front plate **202**. For example, a front plate **202** may be disposed on one side (e.g., a +z direction) of the display **201** and a frame structure **240** may be disposed on the other side (e.g., a −z direction).
[0076] According to an embodiment, the front plate **202** may be coupled to the display **201**. For example, the front plate **202** and the display **201** may adhere to each other through an optical adhesive member (e.g., optically clear adhesive (OCA) or optically clear resin (OCR)) interposed therebetween.
[0077] According to an embodiment, the front plate **202** may be coupled to the frame structure **240**. For example, the front plate **202** may include an outside portion extending outside the display **201** when viewed in the z-axis direction, and may adhere to the frame structure **240** through an adhesive member (e.g., a double-sided tape) disposed between the outside portion of the front plate **202** and the frame structure **240** (e.g., the sidewall **241**). However, it is not limited by the above-described example.
[0078] According to an embodiment, the first printed circuit board **250** and/or the second printed circuit board **252** may be equipped with a processor (e.g., a processor **120** of FIG. **1**), a memory (e.g., a memory **130** of FIG. **1**), and/or an interface (e.g., an interface **177** of FIG. **1**). The processor may include, for example, one or more of a central processing unit, an application processor, a graphic processing unit, an image signal processor, a sensor hub processor, or a communication processor. The memory may include, for example, a volatile memory or a nonvolatile memory. The interface may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, an SD card interface, and/or an audio interface. The interface may electrically or physically connect the electronic device **101** to an external electronic device, and may include a USB connector, an SD card/MMC connector, or an audio connector. According to an embodiment, the first printed circuit board **250** and the second printed circuit board **252** may be operatively or electrically connected to each other through a connection member (e.g., a flexible printed circuit board).
[0079] According to an embodiment, the battery **270** may supply power to at least one component of the electronic device **101**. For example, the battery **270** may include a rechargeable secondary cell or a fuel cell. At least a portion of the battery **270** may be disposed on substantially the same plane as the first printed circuit board **250** and/or the second printed circuit board **252**.
[0080] The electronic device **101** according to an embodiment may include an antenna module (not illustrated) (e.g., an antenna module **197** of FIG. **1**). According to an embodiment, the antenna module may be disposed between the rear plate **211** and the battery **270**. The antenna module may include, for example, a near field communication (NFC) antenna, a wireless charging antenna, and/or a magnetic secure transmission (MST) antenna. The antenna module, for example, may perform short-range communication with an external device, or wirelessly transmit and receive power to and from the external device.

[0081] According to an embodiment, the front camera **205** may be disposed in at least a portion (e.g., a support portion **243**) of the frame structure **240** so that the lens may receive external light through a partial area (e.g., a camera area **237**) of the front plate **202**.
[0082] According to an embodiment, the rear camera **212** may be disposed between the frame structure **240** and the rear plate **211**. According to an embodiment, the rear camera **212** may be electrically connected to the first printed circuit board **250** through a connection member (e.g., a connector). According to an embodiment, the rear camera **212** may be disposed such that the lens may receive external light through a camera area **284** of the rear plate **211** of the electronic device **101**.
[0083] According to an embodiment, the camera area **284** may be formed on the surface (e.g., a rear surface **200**B of FIG. **2**) of the rear plate **211**. According to an embodiment, the camera area **284** may be formed to be at least partially transparent so that external light may be incident to the lens of the rear camera **212**. According to an embodiment, at least a portion of the camera area **284** may protrude from the surface of the rear plate **211** to a predetermined height. However, it is not limited to thereto, and in another embodiment, the camera area **284** may form a plane substantially the same as the surface of the rear plate **211**.
[0084] According to an embodiment, the housing (e.g., a housing **210** of FIG. **2**) of the electronic device **101** may mean a configuration or structure forming at least a portion of the exterior of the electronic device **101**. In this regard, at least a portion of the front plate **202**, the frame structure **240**, and/or the rear plate **211** forming the exterior of the electronic device **101** may be referred to as the housing of the electronic device **101**.
[0085] FIG. **4**A illustrates an exemplary electronic device. FIG. **4**B is a cross-sectional view of an exemplary vapor chamber cut along line A-A′ of FIG. **4**A.
[0086] Referring to FIGS. **4**A and **4**B, an electronic device **101** may include a first printed circuit board **250**, a processor **120**, a vapor chamber **400**, and a metal plate **300**.
[0087] According to an embodiment, the processor **120** may be disposed on the first printed circuit board **250**. The processor **120** may be electrically connected to the first printed circuit board **250**. The processor **120** may be disposed on one surface of the first printed circuit board **250** facing a display (e.g., a display **201** of FIG. **3**). The processor **120** may perform data processing or calculation as the electronic device **101** operates. The processor **120** may emit heat according to performing of data processing or calculation.
[0088] According to an embodiment, the vapor chamber **400** may be spaced apart from the first printed circuit board **250** and at least partially disposed over the processor **120**.
[0089] For example, at least a portion (e.g., a frame structure **240**) of the housing **210** of the electronic device **101** may be disposed on one surface of the first printed circuit board **250** facing the +z direction. The processor **120** may be mounted on the one surface of the first printed circuit board **250**. The vapor chamber **400** may be spaced apart from the first printed circuit board **250** by the at least a portion of the housing **210**. The vapor chamber **400** may be disposed above the processor **120** based on the +z direction. For example, the at least a portion of the housing **210** may be disposed between the processor **120** and the vapor chamber **400**. Since the vapor chamber **400** overlaps the processor **120** when viewed from above, at least some of the heat emitted from the processor **120** may be transferred to the vapor chamber **400**. The vapor chamber **400** may disperse the heat transferred from the processor **120** in the +y direction and/or the −y direction.
[0090] The vapor chamber **400** has been described as dispersing heat of the processor **120**, but is not limited thereto. By at least partially overlapping at least one electronic component that emits heat as the electronic device **101** operates, including the first printed circuit board **250**, the vapor chamber **400** may be configured to disperse heat emitted from the at least one electronic component.
[0091] According to an embodiment, the metal plate **300** may support the vapor chamber **400**. The metal plate **300** may be disposed on the first printed circuit board **250**. For example, the metal plate

300 may be an internal structure (e.g., the frame structure **240** of FIG. **3**) of the housing **210**. For example, the metal plate **300** may include a portion disposed between the processor **120** disposed on one surface of the first printed circuit board **250** facing the metal plate **300** and the vapor chamber **400**. Since the metal plate **300** contacts at least a portion of the vapor chamber **400**, the heat emitted from the processor **120** and transferred to the vapor chamber **400** may diffuse to the metal plate **300** contacting the vapor chamber **400**. The metal plate **300** may include at least one of aluminum and stainless steel, but is not limited thereto. By including the metal plate **300**, the electronic device **101** may be configured to prevent the vapor chamber **400** from being directly exposed to the heat emitted from the processor **120** and to disperse the heat transferred from the processor **120** to the vapor chamber **400** through the metal plate **300**.

[0092] According to an embodiment, the metal plate **300** may include a seating groove **310** in which at least a portion of the vapor chamber **400** is accommodated. The seating groove **310** may include a first portion **311** on the processor **120** for transferring the heat emitted from the processor **120** to the vapor chamber **400**, and a second portion **312** extending from the first portion **311** and having a slit **312***a* formed thereon.

[0093] For example, the seating groove **310** may protrude from the metal plate **300** toward the −z direction in order to accommodate the vapor chamber **400**. For example, the seating groove **310** may cover at least a portion of the vapor chamber **400**. For example, the seating groove **310** may include supporting structures **311***a* and **312***b* configured so that the vapor chamber **400** may be supported on the metal plate **300**.

[0094] For example, when viewing the metal plate **300** from above, the first portion **311** may at least partially overlap the processor **120**. The first portion **311** may include a first supporting structure **311***a* configured to support the vapor chamber **400** and space the processor **120** and the vapor chamber **400** apart. The first supporting structure **311***a* may be disposed on a path through which the heat emitted from the processor **120** is transferred to the vapor chamber **400**. For example, the first supporting structure **311***a* may be referred to as a supporting portion **243** of FIG. **3**, but is not limited thereto.

[0095] For example, since the second portion **312** includes the slit **312***a*, it may emit heat transferred from the vapor chamber **400** through the slit **312***a*. For example, the slit **312***a* may be disposed under the vapor chamber **400**. A portion of the vapor chamber **400** may face a component (e.g., a battery **270** of FIG. **3**) of the electronic device **101** disposed under the slit **312***a* through the slit **312***a*. For example, the second portion **312** may include a second supporting structure **312***b* supporting the vapor chamber **400** so that the vapor chamber **400** does not pass through the slit **312***a*. Regarding that the second supporting structure **312***b* supports the vapor chamber **400**, it will be described later with reference to FIGS. **7**A to **7**C.

[0096] For example, the heat emitted from the processor **120** may be transferred to the vapor chamber **400** in the +z direction through the first portion **311** of the seating groove **310**. The heat transferred to the vapor chamber **400** may be diffused in the −y direction within the vapor chamber **400**. The heat diffused in the −y direction may be emitted through the slit **312***a* of the second portion **312** in the −z direction. However, it is not limited thereto. By including the seating groove **310** accommodating the vapor chamber **400**, the electronic device **101** may provide an additional space in which another electronic component may be mounted inside the electronic device **101** and may transfer the heat emitted from the processor **120** to the vapor chamber **400**.

[0097] According to an embodiment, the vapor chamber **400** may include a first plate **410**. The first plate **410** may include a base **411** supported by the first portion **311** of the seating groove **310** and at least a portion thereof disposed on the slit **312***a*, and a sidewall **412** extending from a periphery of the base **411** toward the outside of the seating groove **310**.

[0098] For example, the base **411** may be supported by the first supporting structure **311***a* of the first portion **311**. Since the base **411** contacts the first portion **311**, the heat emitted from the processor **120** may be transferred through the first portion **311**. For example, a portion of the base

411 may contact the first supporting structure **311***a* of the first portion **311**, and a remaining portion of the base **411** may face the slit **312***a* of the second portion **312**. The remaining portion of the base **411** may be exposed to the outside of the seating groove **310** through the slit **312***a*. The portion of the base **411** may receive heat from the processor **120** through the first supporting structure **311***a*. The remaining portion of the base **411** may emit heat from the vapor chamber **400** through the slit **312***a*. For example, the sidewall **412** may be connected to the base **411**. The side wall **412** may face an inner surface of the seating groove **310**. A height of the sidewall **412** may be substantially the same as a depth of the seating groove **310**.

[0099] The first plate **410** has been described as including the base **411** supported by the first portion **311** of the seating groove **310** and the sidewall **412** extending from the periphery of the base **411**, but is not limited thereto. For example, the first plate **410** may be the base **411** supported by the first portion **311**. The second plate **420** contacting the first plate **410** may be configured to form an inner space S of the vapor chamber **400**, by including a portion extending from at least a portion of the second plate **420** to the first plate **410** and supported by the first plate **410**. For example, the first plate **410** may be the base **411** spaced apart from the second plate **420**. The sidewall **412** may be a separate plate that surrounds a space between the first plate **410** and the second plate **420** and extends from the first plate **410** to the second plate **420**. However, it is not limited thereto, and the vapor chamber **400** may include a structure forming the inner space S in the seating groove **310**.

[0100] According to an embodiment, the vapor chamber **400** may include a second plate **420** contacting a portion of the sidewall **412** of the first plate **410**. The second plate **420** may be coupled to the first plate **410**. The second plate **420** may cover the seating groove **310**.

[0101] According to an embodiment, the second plate **420** may include a first region **421** disposed on the first portion **311** of the seating groove **310**, a second region **422** disposed on the slit **312***a* of the second portion **312** extending from the first portion **311**, and a third region **423** extending from the first region **421** to the second region **422** and located on the boundary b between the first portion **311** and the second portion **312**. In the present document, when an element is referred to as being “on” another element, it should be understood that it is directly on the other element or intervening elements may exist therebetween. For example, in the present document, “B disposed on A” may indicate “B disposed over A”. For example, in the present document, “B disposed on A” may indicate “B faced away from A”.

[0102] For example, when viewing the vapor chamber **400** from above, the first region **421** may overlap the first portion **311** and/or the processor **120**. The second region **422** may overlap the slit **312***a* of the second portion **312**. The third region **423** may overlap the boundary b between the first portion **311** and the second portion **312**.

[0103] For example, the slit **312***a* may extend from the first supporting structure **311***a* of the first portion **311** supporting at least a portion of the base **411**. The boundary b between the first portion **311** and the second portion **312** may include a boundary between the first supporting structure **311***a* and the slit **312***a*. The third region **423** may be a region disposed on the boundary between the first supporting structure **311***a* and the slit **312***a*.

[0104] According to an embodiment, the second plate **420** may include a plurality of pillars **430** protruding toward the base **411** of the first plate **410** and including an end **430***a* having a flat shape at a position adjacent to the base **411**. For example, each of the plurality of pillars **430** may have a flat shape parallel to the base **411** at the position adjacent to the base **411**. For example, each of the plurality of pillars **430** may include a dome shape, but is not limited thereto. For example, when the electronic device **101** receives an impact from the outside, the second plate **420** may temporarily move toward the base **411**. Since the plurality of pillars **430** have a flat shape at the position adjacent to the base **411**, when the second plate **420** moves, the plurality of pillars **430** may increase a supporting area for supporting the second plate **420**. The plurality of pillars **430** may reduce damage (e.g., folding) of the second plate **420** due to external impact of the electronic device **101** by increasing the supporting area. The plurality of pillars **430** may be formed through a press process of the second plate **420**, but is not limited thereto.

[0105] According to an embodiment, the vapor chamber may further include an inner space S including a fluid (not illustrated) covered by the first plate **410** and the second plate **420**, and a wick **450** for moving the fluid in a liquid state and contacting the base **411** in the inner space S. The first plate **410** may be configured to seal the space S together with the second plate **420**. The fluid disposed in the inner space S of the vapor chamber **400** may be vaporized by the heat transferred from the processor **120**. According to an embodiment, the vapor chamber **400** may include at least one of stainless steel and titanium, but is not limited thereto.

[0106] For example, the air pressure of the inner space S of the vapor chamber **400** may be lower than atmospheric pressure. For example, the air pressure of the inner space S of the vapor chamber **400** may be substantially in a vacuum state. The fluid may be, for example, water, but is not limited thereto. For example, the wick **450** may be supported by the base **411** in the inner space S. The wick **450** may transfer the fluid in the liquid state from the inner space S to a portion close to the processor **120**. For example, the wick **450** may include a porous structure for absorbing fluid in the vapor chamber **400**, but is not limited thereto. For example, when the heat from the processor **120** is transferred to the wick **450** through the base **411**, the fluid in the liquid state in the wick **450** may be vaporized. As the fluid vaporized from the wick **450** diffuses into the inner space S of the vapor chamber **400**, the heat from the processor **120** may be dispersed. For example, as the heat from the vapor chamber **400** is emitted through the slit **312***a* of the seating groove **310**, the fluid adjacent to the slit **312***a* may be cooled. The fluid liquefied by cooling may be absorbed by the wick **450**.

[0107] For example, a portion of the sidewall **412** of the first plate **410** may seal the inner space S of the vapor chamber **400**, by contacting the second plate **420**. The sidewall **412** may seal the inner space S by being welded to the second plate **420**, but is not limited thereto.

[0108] According to an embodiment, the second plate **420** may include the plurality of pillars **430** including the end **430***a* protruding toward the base **411** of the first plate **410** and having the flat shape at the position adjacent to the base **411**. The plurality of pillars **430** may include a first set of pillars **431** having a first pattern p**1** in the first region **421** and the second region **422** and a second set of pillars **432** having a second pattern p**2** different from the first pattern p**1** in the third region **423**. For example, each of the first region **421**, the second region **422**, and the third region **423** may include a plurality of pillars having a flat shape at positions adjacent to the base **411**. The pillars included in each of the first region **421** and the second region **422** may be disposed to have the first pattern p**1**. The pillars included in the third region **423** may be disposed to have the second pattern p**2** different from the first pattern p**1**. Regarding that the second pattern p**2** is different from the first pattern p**1**, it will be described later with reference to FIGS. **5**A to **6**B. Since the second pattern p**2** is different from the first pattern p**1**, the vapor chamber **400** may reduce damage due to external impact of the third region **423** disposed on the boundary b between the first portion **311** and the second portion **312**. According to an embodiment, a rigidity of the third region **423** may be greater than a rigidity of the first region **421** and a rigidity of the second region **422**. For example, the second set of pillars **432** included in the third region **423** may be configured such that the rigidity of the third region **423** is greater than the rigidity of the first region **421** and the second region **422** by having the second pattern p**2** different from the first pattern p**1** of the first set of pillars **431** included in the first region **421** and the second region **422**.

[0109] According to the above-described embodiment, the electronic device **101** may reduce damage of the electronic device **101** due to the heat, by including the vapor chamber **400** dispersing the heat emitted from the processor **120** in the electronic device **10**. The electronic device **101** may provide additional space for electronic components within the electronic device **101** and may help heat transfer from the vapor chamber **400**, by including the seating groove **310** accommodating at least a portion of the vapor chamber **400**. The second plate **420** of the vapor chamber **400** may reduce damage of the second plate **420** due to the external impact, by including the plurality of pillars **430** having the flat shape at the position adjacent to the base **411** of the first plate **410**. The second set of pillars **432** in the third region **423** of the second plate **420** may reduce damage of the third region **423** due to the external impact, by having a different pattern from the first set of pillars **431** in the first region **421** and the third region **423**.

[0110] FIGS. **5**A, **5**B, and **5**C illustrate a portion of an exemplary vapor chamber.

[0111] Referring to FIGS. **5**A, **5**B, and **5**C, a second plate **420** of a vapor chamber **400** may include a first region **421**, a second region **422**, a third region **423**, and a plurality of pillars **430**. The plurality of pillars **430** may include a first set of pillars **431** having a first pattern p**1** in the first region **421** and the second region **422** and a second set of pillars **432** having a second pattern p**2** different from the first pattern p**1** in the third region **423**. According to an embodiment, the first pattern p**1** may have a shape in which each of the pillars included in the first set of pillars **431** within the first region **421** and the second region **422** is spaced apart from each other in a first distance d**1**. According to an embodiment, the first distance d**1** may be located within a range of 2 mm or more and 4 mm or less, but is not limited thereto.

[0112] Hereinafter, redundant descriptions of the configurations described in FIGS. **4**A to **4**B will be omitted.

[0113] Referring to FIGS. **5**A and **5**B, the second pattern p**2** may include at least one bead **510** to which at least a portion of the second set of pillars **432** is connected. For example, the at least one bead **510** may have a shape in which the at least a portion of the second set of pillars **432** in the third region **423** are connected to each other in an inner space (e.g., an inner space S of FIG. **4**B) of the vapor chamber **400**. For example, the at least one bead **510** may have a structure in which the at least a portion of the second set of pillars **432** are connected to each other through rivet bonding, but is not limited thereto. The third region **423** may reduce damage of the third region **423** due to external impact, by including the at least one bead **510** to which the at least a portion of the second set of pillars **432** are connected to the second pattern p**2**.

[0114] According to an embodiment, the at least one bead **510** may include a plurality of beads each having a length in the first direction (e.g., −y direction). The second pattern p**2** may include a first set of beads **511** arranged in a second direction (e.g., +x direction) perpendicular to the first direction. The at least one bead **510** may help the fluid move in the vapor chamber **400** in the first direction by including the plurality of beads each having the length in the first direction, and may reduce the damage of the third region **423** from the external impact by including the first set of beads **511** arranged in the second direction perpendicular to the first direction. According to an embodiment, the third region **423** may include pillars each disposed between the beads included in the first set of beads **511**. The pillars respectively disposed between the beads included in the first set of beads **511** may provide a movement path of fluid in the vapor chamber **400**, by reinforcing a region between the beads and being spaced apart from each of the beads.

[0115] Referring to FIG. **5**A, the third region **423** may include a second set of beads **512** and a third set of beads **513** spaced apart from the first set of beads **511**. The first set of beads **511**, the second set of beads **512**, and the third set of beads **513** may form a pattern of a zigzag shape. Through the zigzag shape, the third region **423** may reduce the damage of the third region **423** due to the external impact.

[0116] According to an embodiment, the first pattern p**1** may have a shape in which each of the first set of pillars **431** is spaced apart from each other in the first distance d**1**, and the at least one bead **510** may include two to eight pillars spaced apart from each other in the first distance d**1** in the first direction (e.g., −y direction) of the second set of pillars **432**. For example, referring to FIG. **5**A, the at least one bead **510** may include two pillars spaced apart from each other in the first distance d**1**. The second pattern p**2** may include the first set of beads **511** having the length in the first direction. For example, referring to FIG. **5**B, the at least one bead **510** may include eight pillars spaced apart from each other in the first distance d**1**. The second pattern p**2** may include a fourth set of beads **514** having the length in the first direction. Each of the beads included in the fourth set of beads

514 may have a length longer than the length of each of the beads included in the first set of beads **511**, by including more pillars than each of the beads included in the first set of beads **511** of FIG. **5**A. Each of the beads included in the fourth set of beads **514** may have a shape in which pillars spaced apart from each other, respectively, in a first distance d**1** in the first direction are connected to each other. Each of the beads included in the fourth set of beads **514** may reduce the damage of the third region **423** from the external impact, by having the length longer than the length of each of the beads included in the first set of beads **511**. By including the at least one bead **510** in a form in which the second pattern p**2** includes two to eight pillars having the length in the first direction and spaced apart from each other in the first distance d**1**, the third region **423** may help the fluid move in the vapor chamber **400** in the first direction and may reduce the damage of the third region **423** due to the external impact.

[0117] Referring to FIG. **5**C, the second plate **420** may further include a third pattern p**3** including the at least one bead **510** connected to at least a portion of a plurality of pillars **430** disposed along a periphery of the second plate **420**. According to an embodiment, the third pattern p**3** may include a plurality of beads and at least one pillar **520** each disposed between the plurality of beads. The third pattern p**3** may reduce occurrence of cracks at the periphery of the second plate **420** due to the external impact, by including the at least one bead **510** disposed along the periphery of the second plate **420**. Since the fluid in the vapor chamber **400** is a fountain flow, the at least one bead **510** may reduce damage to the periphery of the second plate **420** without reducing heat diffusion by the fluid by being disposed on the periphery of the second plate **420**. The third pattern p**3** may be configured to emit heat from the vapor chamber **400** between the plurality of beads, by including the at least one pillar **520** each disposed between the plurality of beads. According to an embodiment, the third pattern p**3** may be located in the first region **421** and/or the second region **422**. According to an embodiment, the third pattern p**3** may be located in the first region **421**, the second region **422**, and/or the third region **423**.

[0118] According to the above-described embodiment, the vapor chamber **400** may reduce damage of the third region **423** due to the external impact, by including the at least one bead **510** in which at least a portion of the second set of pillars **432** are connected to each other in the third region **423** of the second plate **420**. The third region **423** may help the fluid move, by having a length in the direction (e.g., −y direction) in which the fluid in the vapor chamber **400** moves.

[0119] FIGS. **6**A and **6**B illustrate a portion of an exemplary vapor chamber.

[0120] Referring to FIGS. **6**A and **6**B, a second plate **420** of a vapor chamber **400** may include a first region **421**, a second region **422**, a third region **423**, and a plurality of pillars **430**. The plurality of pillars **430** may include a first set of pillars **431** having a first pattern p**1** in the first region **421** and the second region **422** and a second set of pillars **432** having a second pattern p**2** different from the first pattern p**1** in the third region **423**. The first pattern p**1** may have a shape in which each of the first set of pillars **431** is spaced apart from each other in a first distance d**1**.

[0121] According to an embodiment, the second pattern p**2** may have a shape in which each of the pillars included in the second set of pillars **432** is spaced apart from each other at various distances. The second pattern p**2** may reduce damage (e.g., folding) of the third region **423** due to external impact, by including the shape in which each of the pillars included in the second set of pillars **432** is spaced apart from each other at various distances. For example, unlike the first set of pillars **431** having the first pattern p**1** is regularly arranged, the second set of pillars **432** having the second pattern p**2** may be arranged irregularly.

[0122] For example, referring to FIG. **6**A, the second pattern p**2** may have a shape in which one pillar **632***a* among the second set of pillars **432** is spaced apart from another pillar **632***b* most adjacent to the one pillar **632***a* in a second distance d**2** smaller than the first distance d**1**. For example, the number of pillars per unit area of the second set of pillars **432** having the second pattern p**2** may be greater than the number of pillars per unit area of the first set of pillars **431** having the first pattern p**1**. Since the second pattern p**2** has the shape in which one pillar **632***a* among the second set of pillars **432** is spaced apart from another pillar **632***b* most adjacent to the one pillar **632***a* in the second distance d**2** smaller than the first distance d**1**, the third region **423** may reduce damage of the third region **423** from the external impact.

[0123] For example, referring to FIG. **6**B, the second pattern p**2** may have a shape in which one pillar **632***c* among the second set of pillars **432** is spaced apart in a third distance d**3** greater than the first distance d**1** from the farthest other one pillar **632***d* among pillars surrounding the one pillar **632***c*. For example, the number of pillars per unit area of the second set of pillars **432** having the second pattern p**2** may be less than the number of pillars per unit area of the first set of pillars **431** having the first pattern p**1**. For example, the third region **423** may be a flow bottleneck in a path through which the fluid in the vapor chamber **400** moves. Since the second pattern p**2** has the shape in which one pillar **632***c* among the second set of pillars **432** is spaced apart in the third distance d**3** greater than the first distance d**1** from the farthest other one pillar **632***d* among pillars surrounding the one pillar **632***c*, the third region **423** may help fluid in the vapor chamber **400** move.

[0124] According to an embodiment, the second set of pillars **432** may include a third set of pillars **433** spaced apart from each other in the first distance d**1** in the first direction (e.g., −y direction) in the second pattern p**2**, and a fourth set of pillars **434** disposed in a second direction (e.g., +x direction) perpendicular to the first direction with respect to the third set of pillars **433** and spaced apart from each other in the first direction in the third distance d**3** greater than the first distance d**1**, in the second pattern p**2**. As the pillars included in the third set of pillars **433** and the pillars included in the fourth set of pillars **434** are spaced apart from each other, respectively, in different distances, the third region **423** may reduce bending of the third region **423** due to the external impact and may help the movement of the fluid in the vapor chamber **400**.

[0125] According to the above-described embodiment, the vapor chamber **400** may reduce the damage of the third region **423** from the external impact as the second pattern p**2** included in the third region **423** is different from the first pattern p**1** included in each of the first region **421** and the second region **422**, or may help the movement of the fluid in the vapor chamber **400** moving through the third region **423** by including irregularly arranged pillars.

[0126] FIG. **7**A illustrates the inside of an exemplary electronic device in which a vapor chamber is disposed. FIG. **7**B is a cross-sectional view of an exemplary electronic device cut along line B-B′ of FIG. **7**A. FIG. **7**C is a cross-sectional view of an exemplary electronic device cut along line C-C′ of FIG. **7**A.

[0127] Referring to FIGS. **7**A, **7**B, and **7**C, an electronic device **101** may be spaced apart from a printed circuit board **250**, a processor **120** on the printed circuit board **250**, and a vapor chamber **400** spaced apart from the printed circuit board **250** and at least partially overlapping the processor **120** when viewed from above. The electronic device **101** may include a metal plate **300** on the printed circuit board **250** supporting the vapor chamber **400**. The metal plate **300** may include a first portion **311** on the processor **120** for transferring heat emitted from the processor **120** to the vapor chamber **400** and a second portion **312** extending from the first portion **311** and having a slit **312***a* formed therein, and may include a seating groove **310** in which at least a portion of the vapor chamber **400** is accommodated. The vapor chamber **400** may include a first plate **410** and a second plate **420**. The first plate **410** may include a base **411** supported by the first portion **311** of the seating groove **310** and having at least a portion disposed on the slit **312***a* and a sidewall **412** extending from a periphery of the base **411** toward the outside of the seating groove **310**. The second plate **420** may include a first region **421** contacting the sidewall **412** and disposed on the first portion **311**, a second region **422** disposed on the slit **312***a* of the second portion **312**, a third region **423** extending from the first region **421** to the second region **422** and located on the boundary b between the first region **311** and the second region **312**, and a plurality of pillars **430** protruding toward the base **411** and including an end having a flat shape at a position adjacent to the base **411**. The plurality of pillars **430** may include a first set of pillars **431** having a first pattern p**1** in the first region **421** and the second region **422** and a second set of pillars **432** having a second pattern p**2** different from the first pattern p**1** in the third region **423**.

[0128] It has been described that the first region **421** and the second region **422** include the first set of pillars **431** having the first pattern p**1**, but it is not limited thereto. The pillars included in the first region **421** may have a different pattern from the pillars included in the second region **422**. The pillars included in the third region **423** may be configured to reduce damage of the third region **423** due to external impact by having a pattern different from the pattern of the pillars included in the first region **421** and the pattern of the pillars included in the second region **422**.

[0129] According to an embodiment, the electronic device **101** may further include a display **201**. The vapor chamber **400** may be disposed under the display **201**. One surface of the metal plate **300** on which the seating groove **310** accommodating the at least a portion of the vapor chamber **400** is formed may face the display **201**. The electronic device **101** may reduce damage to the display **201** due to the heat emitted from the processor **120**, by including the vapor chamber **400** and the metal plate **300** between the processor **120** and the display **201**.

[0130] According to an embodiment, the first portion **311** of the seating groove **310** may include a first supporting structure **311***a* supporting the first plate **410**. The second portion **312** of the seating groove **310** may include a second supporting structure **312***b* protruding from an inner surface of the seating groove **310** and disposed along a periphery of the slit **312***a*. The vapor chamber **400** may further include a flange portion **460** disposed along a periphery of the vapor chamber **400** and supporting the vapor chamber **400** through the second supporting structure **312***b*.

[0131] For example, the first supporting structure **311***a* may support the first plate **410** including the base **411** by supporting at least a portion of the base **411**. For example, the first supporting structure **311***a* may be a path through which the heat emitted from the processor **120** is transferred to the first plate **410**. For example, the second supporting structure **312***b* may extend from the first supporting structure **311***a* and/or the periphery of the slit **312***a*. For example, the second supporting structure **312***b* may be configured so that the vapor chamber **400** may be supported by the metal plate **300** without being separated from the seating groove **310** by the slit **312***a*, by supporting a portion of the sidewall **412** of the first plate **410** and/or a portion of the second plate **420**.

[0132] For example, the flange portion **460** may be a portion of the vapor chamber **400** supported by the second supporting structure **312***b*. For example, the flange portion **460** may be a portion overlapping the second supporting structure **312***b* when viewing the metal plate **300** from above. For example, the flange portion **460** may be a portion extending from a periphery of the second plate **420** and/or a periphery of the first plate **410** toward the outside of the seating groove **310**. For example, the flange portion **460** may be a portion extending in a direction parallel to the direction in which the second plate **420** extends from the periphery of the vapor chamber **400**. The vapor chamber **400** may be configured such that the vapor chamber **400** is supported by the metal plate **300** in the second region **321** including the slit **312***a*, by including the flange portion **460** supporting the vapor chamber **400** through the second supporting structure **312**b.

[0133] According to an embodiment, a width w**1** of the base **411** disposed on the first portion **311** may be greater than a width w**2** of the base **411** disposed on the second portion **312**. For example, since a portion of the vapor chamber **400** disposed in the first portion **311** is supported by the first supporting structure **311***a*, it may not include the flange portion **460**. Since the portion of the vapor chamber **400** does not include the flange portion **460**, a width w**3** of the first portion **311** for accommodating the portion of the vapor chamber **400** may be greater than a width w**4** of the second portion **312** including the second supporting structure **312***b*. Since the width w**3** of the first portion **311** of the seating groove **310** accommodating the vapor chamber **400** is greater than the width w**4** of the second portion **312**, the width w**1** of the base **411** disposed on the first portion **311** may be greater than the width w**2** of the base **411** disposed on the second portion **312**.

[0134] For example, a portion of the vapor chamber **400** disposed on the first portion **311** may increase an inner space S of the vapor chamber **400**, by not including the flange portion **460**. A width of a wick **450** contacting the base **411** disposed on the first portion **311** may be greater than that of the case including the flange portion **460**. The first set of pillars **431** disposed on the first region **421** of the second plate **420** may increase as the inner space S of the vapor chamber **400** increases. As the inner space S of the vapor chamber **400** increases, the vapor chamber **400** may provide an additional space capable of operating to the fluid in the inner space S. The vapor chamber **400** may increase the diffusion speed of the heat transferred from the processor **120** by providing the additional space.

[0135] According to an embodiment, the flange portion **460** may include a first protrusion **461** extending from at least a portion of the second region **422** and the third region **423** of the second plate **420**, and a second protrusion **462** extending from a periphery of the sidewall **412** of the first plate **410** and supporting the first protrusion **461**. For example, the first protrusion **461** may extend in the direction parallel to the direction in which the second plate **420** extends. The second protrusion **462** may extend along the first protrusion **461** in the direction parallel to the direction in which the second plate **420** extends. The sidewall **412** may be configured to support the second plate **420** and to seal the inner space S of the vapor chamber **400**, by including the second protrusion **462**. The second plate **420** may increase a contact surface of the flange portion **460** with the second supporting structure **312***b*, by including the first protrusion **461**.

[0136] It was described that the first portion **311** disposed on the processor **120** includes the first supporting structure **311***a* supporting the first plate **410** and the second portion **312** extending from the first portion **311** includes the slit **312***a*, thereby providing the second supporting structure **312***b* supporting the flange portion **460**, but is not limited thereto.

[0137] Although not illustrated, unlike those illustrated in FIGS. **7**B and **7**C, the first portion **311** may include a slit configured such that at least a portion (e.g., the base **411**) of the first plate **410** faces the processor **120**. The first portion **311** may further include a heat transfer material disposed in the slit for heat transfer from the processor **120** to the vapor chamber **400**. The vapor chamber **400** disposed on the first portion **311** may be supported by the first portion **311** by including a flange portion. Unlike those illustrated in FIGS. **7**B and **7**C, the second portion **312** extending from the first portion **311** may support at least a portion of the first plate **410**, by not including the slit **312***a*. The second portion **312** may provide an additional space in the inner space S of the vapor chamber **400**, by not including the flange portion. Heat transferred from the processor **120** to the vapor chamber **400** through the slit may reduce damage to electronic components (e.g., the display **201**) disposed on the vapor chamber **400**, by diffusing to the metal plate **300** by the vapor chamber **400**. However, it is not limited thereto.

[0138] According to an embodiment, the electronic device **101** may further include a first adhesive member **711** disposed between the base **411** and the first supporting structure **311***a*, and a second adhesive member **712** disposed between the flange portion **460** and the second supporting structure **312***b*. By including the first adhesive member **711** and the second adhesive member **712**, the electronic device **101** may fasten the vapor chamber **400** to the metal plate **300** and may reduce separation of the vapor chamber **400** from the seating groove **310**.

[0139] According to an embodiment, the first adhesive member **711** and/or the second adhesive member **712** may include a material having high thermal conductivity for heat transfer between the vapor chamber **400** and the metal plate **300**. For example, it may include at least one of a thermal tape and a thermal pad, but is not limited thereto.

[0140] According to the above-described embodiment, the electronic device **101** may reduce separation of the vapor chamber **400** from the metal plate **300**, by including the first supporting structure **311***a* and the second supporting structure **312***b*. The vapor chamber **400** may be configured such that the vapor chamber **400** is not separated through the slit **312***a*, by including the flange portion **460** supported by the second supporting structure **312***b*. The vapor chamber **400** may increase the inner space S in which the fluid in the vapor chamber **400** operates without including the flange portion **460** in a portion of the vapor chamber **400** disposed on the first portion **311** of the settling groove **310**.

[0141] According to the above-described embodiment, an electronic device (e.g., an electronic device **101** of FIG. **1**) may comprise a printed circuit board (e.g., a first printed circuit board **250** of FIG. **3**), a processor (e.g., a processor **120** of FIG. **1**) on the printed circuit board, and a vapor chamber (e.g., a vapor chamber **400** of FIG. **4**A) spaced apart from the printed circuit board and at least partially disposed over the processor. The electronic device may comprise a metal plate (e.g., a metal plate **300** of FIG. **4**A), disposed on the printed circuit board, including a seating groove including a first portion (e.g., a first portion **311** of FIG. **4**A) on the processor for supporting the vapor chamber and transferring heat emitted from the processor to the vapor chamber and a second portion (e.g., a second portion **312** of FIG. **4**A) extending from the first portion and having a slit (e.g., a slit **312***a* of FIG. **4**A) formed therein and a seating groove (e.g., a seating groove **310** of FIG. **4**A) in which at least a portion of the vapor chamber is accommodated. The vapor chamber may include a first plate (e.g., a first plate **410** of FIG. **4**A) including a base (e.g., a base **411** of FIG. **4**B) supported by the first portion of the seating groove and having at least a portion disposed on the slit, and a sidewall (e.g., a sidewall **412** of FIG. **4**B) extending from a periphery of the base toward an outside of the seating groove. The vapor chamber may include a second plate (e.g., a second plate **420** of FIG. **4**A) contacting the sidewall and including a first region (e.g., a first region **421** of FIG. **4**A) disposed on the first portion, a second region (e.g., a second region **422** of FIG. **4**A) disposed on the slit of the second portion, a third region (e.g., a third region **423** of FIG. **4**A) extending from the first region to the second region and located on the boundary (e.g., a boundary b of FIG. **4**A) between the first portion and the second portion, and a plurality of pillars (e.g., a plurality of pillars **430** of FIG. **4**A) protruding toward the base and including an end (e.g., an end **430***a* of FIG. **4**B) having flat shape at a position adjacent to the base. The plurality of pillars may include a first set of pillars (e.g., a first set of pillars **431** of FIG. **4**A) having a first pattern (e.g., a first pattern p**1** of FIG. **4**A) in the first region and the second region, and a second set of pillars (e.g., a second set of pillars **432** of FIG. **4**A) having a second pattern (e.g., a second pattern p**2** of FIG. **4**A) different from the first pattern in the third region. According to the above-mentioned embodiment, the electronic device may diffuse heat emitted from the processor by including the vapor chamber. The vapor chamber may reduce damage to the third region due to external impact by including the first set of pillars having the first pattern in the first region and in the second region and the second set of pillars having the second pattern different from the first pattern in the third region. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0142] According to an embodiment, the second pattern may include at least one bead (e.g., at least one bead **510** of FIG. **5**A) to which at least some of the second set of pillars are connected. According to the above-mentioned embodiment, the second pattern may reduce the damage to the third region due to the external impact by including the at least one bead. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0143] According to an embodiment, the at least one bead may include a plurality of beads each having a length in the first direction, and the second pattern may include a set of beads (e.g., a first set of beads **511**, a second set of beads **512**, a third set of beads **513** of FIG. **5**A) arranged in a direction perpendicular to the first direction. According to the above-mentioned embodiment, the second pattern may help movement of the fluid in the vapor chamber moving in the first direction and may reduce the damage to the third region due to the external impact, by including the set of beads arranged in a direction perpendicular to the first direction. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0144] According to an embodiment, the first pattern may have a shape in which each of the first set of pillars spaced apart from each other in a first distance d**1** (e.g., the first distance d**1** of FIG. **5**A), and the at least one bead may include two to eight pillars spaced apart from each other in the first distance in the first direction among the second set of pillars. According to the above-mentioned embodiment, the at least one bead may help the movement of the fluid in the vapor chamber moving in the first direction and may reduce the damage to the third region due to the external impact, by including two to eight pillars spaced apart from each other in the first distance in the first direction among the second set of pillars. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0145] According to an embodiment, the first pattern may have a shape in which each of the first set of pillars spaced apart from each other in the first distance d**1**, and the second pattern may have a shape in which one pillar (e.g., one pillar **632***a* of FIG. **6**A) among the second set of pillars spaced apart in a second distance (e.g., a second distance d**2** of FIG. **6**A) smaller than the first distance from another pillar (e.g., another pillar **632***b* of FIG. **6**A) closest to the one pillar. According to the above-mentioned embodiment, the second pattern may reduce the damage to the third region due to the external impact, by having the shape in which one pillar among the second set of pillars spaced apart in the second distance smaller than the first distance from another pillar closest to the one pillar. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0146] According to an embodiment, the first pattern may have the shape in which each of the first set of pillars spaced apart from each other in the first distance d**1**, and the second pattern may have a shape in which one pillar (e.g., one pillar **632***c* of FIG. **6**B) among the second set of pillars spaced apart in a third distance (e.g., a third distance d**3** of FIG. **6**B) greater than the first distance from another pillar (e.g., another pillar **632***d* of FIG. **6**B) farthest to the one pillar among pillars surrounding the one pillar. According to the above-mentioned embodiment, the second pattern may help the movement of the fluid in the vapor chamber passing through the third region, by having the shape in which one pillar among the second set of pillars spaced apart in the third distance greater than the first distance from another pillar farthest to the one pillar among pillars surrounding the one pillar. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0147] According to an embodiment, the first pattern may have a shape in which each of the first set of pillars spaced apart from each other in the first distance d**1**. The second set of pillars may include a third set of pillars (e.g., a third set of pillars **433** of FIG. **6**B) spaced apart from each other in the first distance in a first direction in the second pattern, and a fourth set of pillars spaced apart from each other in a third distance in the first direction greater than the first distance and disposed in a second direction perpendicular to the first direction with respect to the third set of pillars in the second pattern. According to the above-mentioned embodiment, the second set of pillars may help the movement of the fluid in the vapor chamber passing through the third region, by including the third set of pillars and the fourth set of pillars. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0148] According to an embodiment, a rigidity of the third region may be greater than a rigidity of the first region and the rigidity of the second region. According to the above-mentioned embodiment, as the rigidity of the third region is greater than the rigidity of the first region and the rigidity of the second region, the vapor chamber may reduce the damage to the third region due to the external impact. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0149] According to an embodiment, the second plate may include a third pattern (e.g., a third pattern p**3** of FIG. **5**C) including at least one bead to which at least some of the plurality of pillars disposed along a periphery of the second plate are connected. According to the above-mentioned embodiment, the second plate may reduce damage to the periphery of the second plate due to the external impact, by including the third pattern. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0150] According to an embodiment, the first portion may include a first supporting structure (e.g., a first supporting structure **311***a* of FIG. **4**A) supporting the first plate, and the second portion may include a second supporting structure (e.g., a second supporting structure **312***b* of FIG. **4**B) protruding from an inner surface of the seating groove and disposed along a periphery of the slit. The vapor chamber may further include a flange portion (e.g., a flange portion **460** of FIG. **7**C) disposed along a periphery of the vapor chamber and supporting the vapor chamber through the second supporting structure. According to the above-mentioned embodiment, the seating groove may support the vapor chamber by including the first supporting structure and the second supporting structure. The vapor chamber may reduce separation of the vapor chamber from the metal plate by including the flange portion. A portion of the vapor chamber disposed on the first portion may provide an additional space in a space in which the fluid of the vapor chamber operates by not including the flange portion. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0151] According to an embodiment, the flange portion may include a first protrusion (e.g., a first protrusion **461** of FIG. **7**C) extending from at least a portion of the second region and the third region of the second plate and a second protrusion (e.g., a second protrusion **462** of FIG. **7**C) extending from a periphery of the sidewall of the first plate and supporting the first protrusion. According to the above-mentioned embodiment, the flange portion may reduce the separation of the vapor chamber from the metal plate by including the first protrusion and the second protrusion. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0152] The electronic device according to an embodiment may further comprise a first adhesive member (e.g., a first adhesive member **711** of FIG. **7**C) disposed between the base and the first supporting structure and a second adhesive member (e.g., a second adhesive member **712** of FIG. **7**C) disposed between the flange portion and the second supporting structure. According to the above-mentioned embodiment, the electronic device may reduce separation of the vapor chamber from the seating groove by including the first adhesive member and the second adhesive member. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0153] According to an embodiment, the vapor chamber may further include a space (e.g., an inner space S of FIG. **4**B) covered by the first plate and the second plate and including a fluid, and a wick (e.g., a wick **450** of FIG. **4**B) for moving the fluid in a liquid state and contacting the base in the space. The first plate may be configured to seal the space together with the second plate. According to the above-mentioned embodiment, the vapor chamber may circulate the fluid by including the wick. The first plate may prevent separation of the fluid in the space by sealing the space together with the second plate. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0154] According to an embodiment, a width (e.g., a width w**1** of FIG. **7**B) of the base disposed on the first portion may be greater than a width (e.g., a width w**2** of FIG. **7**C) of the base disposed on the second portion. According to the above-mentioned embodiment, as the width of the base disposed on the first portion is greater than the width of the base disposed on the second portion, the vapor chamber may provide an additional space in the space in which the fluid in the vapor chamber operates. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0155] According to an embodiment, the vapor chamber may further include at least one of stainless steel, and titanium. According to the above-mentioned embodiment, the vapor chamber may increase the diffusion speed of heat transferred to the vapor chamber by including the stainless steel. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0156] According to an embodiment, an electronic device may comprise a printed circuit board, a processor on the printed circuit board, and a vapor chamber spaced apart from the printed circuit board and at least partially disposed over the processor. The electronic device may comprise a metal plate, disposed on the printed circuit board, including a seating groove including a first portion on the processor for supporting the vapor chamber and transferring heat emitted from the processor to the vapor chamber and a second portion extending from the first portion and having a slit formed therein and a seating groove in which at least a portion of the vapor chamber is accommodated. The vapor chamber may include a first plate including a base supported by the first portion of the seating groove and having at least a portion disposed on the slit, and a sidewall extending from a periphery of the base toward an outside of the seating groove. The vapor chamber may include a second plate contacting the sidewall and including a first region disposed on the first portion, a second region disposed on the slit of the second portion, a third region extending from the first region to the second region and located on the boundary between the first portion and the second portion, and a plurality of pillars protruding toward the base and including an end having flat shape at a position adjacent to the base. The vapor chamber may further include at least one of stainless steel and titanium. The plurality of pillars may include a first set of pillars having a first pattern in the first region and the second region, and a second set of pillars having a second pattern different from the first pattern in the third region. A rigidity of the third region may be greater than a rigidity of the first region and a rigidity of the second region. According to the above-mentioned embodiment, the electronic device may diffuse heat emitted from the processor by including the vapor chamber. The vapor chamber may reduce damage to the third region due to external impact by including the first set of pillars having the first pattern in the first region and in the second region and the second set of pillars having the second pattern different from the first pattern in the third region. As the rigidity of the third region is greater than the rigidity of the first region and the rigidity of the second region, the vapor chamber may reduce the damage to the third region due to the external impact The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0157] According to an embodiment, the second pattern may include at least one bead to which at least some of the second set of pillars are connected. According to the above-mentioned embodiment, the second pattern may reduce the damage to the third region due to the external impact by including the at least one bead. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0158] According to an embodiment, the first pattern may have a shape in which each of the first set of pillars spaced apart from each other in the first distance d**1**, and the second pattern may have a shape in which one pillar among the second set of pillars spaced apart in a second distance smaller than the first distance from another pillar closest to the one pillar. According to the above-mentioned embodiment, the second pattern may reduce the damage to the third region due to the external impact, by having the shape in which one pillar among the second set of pillars spaced apart in the second distance smaller than the first distance from another pillar closest to the one pillar. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0159] According to an embodiment, the first portion may include a first supporting structure supporting the first plate, and the second portion may include a second supporting structure protruding from an inner surface of the seating groove and disposed along a periphery of the slit. The vapor chamber may further include a flange portion disposed along a periphery of the vapor chamber and supporting the vapor chamber through the second supporting structure. According to the above-mentioned embodiment, the seating groove may support the vapor chamber by including the first supporting structure and the second supporting structure. The vapor chamber may reduce separation of the vapor chamber from the metal plate by including the flange portion. A portion of the vapor chamber disposed on the first portion may provide an additional space in a space in which the fluid of the vapor chamber operates by not including the flange portion. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0160] According to an embodiment, the vapor chamber may further include a space covered by the first plate and the second plate and including a fluid, and a wick for moving the fluid in a liquid state and contacting the base in the space. The first plate may be configured to seal the space together with the second plate. According to the above-mentioned embodiment, the vapor chamber may circulate the fluid by including the wick. The first plate may prevent separation of the fluid in the space by sealing the space together with the second plate. The above-mentioned embodiment may have various effects including the above-mentioned effects.

[0161] The electronic device according to various embodiments may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0162] It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0163] As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0164] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a complier or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0165] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application

store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.
[0166] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

Claims

1. An electronic device comprising: a printed circuit board; a processor on the printed circuit board; a vapor chamber spaced apart from the printed circuit board and at least partially disposed over the processor; and a metal plate, disposed on the printed circuit board, including a seating groove in which at least a portion of the vapor chamber is accommodated and including a first portion on the processor for transferring heat emitted from the processor to the vapor chamber and a second portion extending from the first portion and having a slit formed therein, wherein, the vapor chamber includes: a first plate including a base supported by the first portion of the seating groove and having at least a portion disposed on the slit, and a sidewall extending from a periphery of the base toward an outside of the seating groove, and a second plate contacting the sidewall and including a first region disposed on the first portion, a second region disposed over the slit of the second portion, a third region extending from the first region to the second region and located on the boundary between the first portion and the second portion, and a plurality of pillars protruding toward the base and including an end having flat shape at a position adjacent to the base, and wherein, the plurality of pillars includes: a first set of pillars having a first pattern in the first region and the second region, and a second set of pillars having a second pattern different from the first pattern in the third region.
2. The electronic device of claim 1, wherein the second pattern includes at least one bead to which at least some of the second set of pillars are connected.
3. The electronic device of claim 2, wherein the at least one bead includes a plurality of beads each having a length in a first direction, and wherein the second pattern includes a set of beads arranged in a second direction perpendicular to the first direction.
4. The electronic device of claim 2, wherein the first pattern has a shape in which each of the first set of pillars spaced apart from each other in a first distance, and wherein, the at least one bead includes two to eight pillars spaced apart from each other in the first distance in the first direction among the second set of pillars.
5. The electronic device of claim 1, wherein the first pattern has a shape in which each of the first set of pillars spaced apart from each other in a first distance, and wherein, the second pattern has a shape in which one pillar among the second set of pillars spaced apart in a second distance smaller than the first distance from another pillar closest to the one pillar.
6. The electronic device of claim 1, wherein the first pattern has a shape in which each of the first set of pillars spaced apart from each other in a first distance, and wherein the second pattern has a

shape in which one pillar among the second set of pillars spaced apart in a third distance greater than the first distance from another pillar farthest to the one pillar among pillars surrounding the one pillar.

7. The electronic device of claim 1, wherein the first pattern has a shape in which each of the first set of pillars spaced apart from each other in a first distance, and wherein the second set of pillars include: a third set of pillars spaced apart from each other in the first distance in a first direction in the second pattern, and a fourth set of pillars spaced apart from each other in a third distance in the first direction greater than the first distance and disposed in a second direction perpendicular to the first direction with respect to the third set of pillars in the second pattern.

8. The electronic device of claim 1, wherein a rigidity of the third region is greater than a rigidity of the first region and the rigidity of the second region.

9. The electronic device of claim 1, wherein the second plate further includes a third pattern including at least one bead to which at least some of the plurality of pillars disposed along a periphery of the second plate are connected.

10. The electronic device of claim 1, wherein the first portion includes a first supporting structure supporting the first plate, wherein the second portion includes a second supporting structure protruding from an inner surface of the seating groove and disposed along a periphery of the slit, and wherein the vapor chamber further includes a flange portion disposed along a periphery of the vapor chamber and supporting the vapor chamber through the second supporting structure.

11. The electronic device of claim 10, wherein the flange portion includes: a first protrusion extending from at least a portion of the second region and the third region of the second plate; and a second protrusion extending from a periphery of the sidewall of the first plate and supporting the first protrusion.

12. The electronic device of claim 10, further comprising: a first adhesive member disposed between the base and the first supporting structure; and a second adhesive member disposed between the flange portion and the second supporting structure.

13. The electronic device of claim 1, wherein the vapor chamber further includes: a space covered by the first plate and the second plate and including a fluid, and a wick for moving the fluid in a liquid state and contacting the base in the space, and wherein the first plate is configured to seal the space together with the second plate.

14. The electronic device of claim 1, wherein a width of the base disposed on the first portion is greater than a width of the base disposed on the second portion.

15. The electronic device of claim 1, wherein the vapor chamber further includes at least one of stainless steel, and titanium.

16. An electronic device comprising: a printed circuit board; a processor on the printed circuit board; a vapor chamber, spaced apart from the printed circuit board, at least partially overlapped the processor when viewed from above; and a metal plate, disposed on the printed circuit board, including a seating groove accommodating at least a portion of the vapor chamber, the seating groove including a first portion on the processor for supporting the vapor chamber and a second portion extending from the first portion and having a slit formed therein, wherein the vapor chamber includes: a first plate including: a base supported by the first portion of the seating groove and at least partially disposed on the slit, and a sidewall extending from a periphery of the base toward an outside of the seating groove, and a second plate being in contact with the sidewall, the second plate including: a first region disposed on the first portion, a second region disposed on the slit of the second portion, a third region extending from the first region to the second region and located on a boundary portion between the first portion and the second portion, and a plurality of pillars protruding toward the base and including an end having flat shape at a position adjacent to the base, wherein the plurality of pillars including: a first set of pillars having a first pattern in the first region and the second region, and a second set of pillars having a second pattern different from the first pattern in the third region, wherein a rigidity of the third region is greater than a rigidity of the first region and a rigidity of the second region, and wherein the vapor chamber includes at least one of stainless steel or titanium.

17. The electronic device of claim 16, wherein the second pattern includes at least one bead to which at least some of the second set of pillars are connected.

18. The electronic device of claim 16, wherein the first pattern has a shape in which each of the first set of pillars spaced apart from each other in the first distance, and wherein the second pattern has a shape in which one pillar from among the second set of pillars spaced apart in a second distance smaller than the first distance from another pillar closest to the one pillar.

19. The electronic device of claim 16, wherein the first portion includes a first supporting structure supporting the first plate, wherein the second portion includes a second supporting structure protruding from an inner surface of the seating groove and disposed along a periphery of the slit, and wherein the vapor chamber includes a flange portion disposed along a periphery of the vapor chamber and supporting the vapor chamber through the second supporting structure.

20. The electronic device of claim 16, wherein the vapor chamber includes: a space covered by the first plate and the second plate and including a fluid, and a wick for moving the fluid in a liquid state and contacting the base in the space, and wherein the first plate is configured to seal the space together with the second plate.
